

F-SERIES

F-8

F-10

F-12

F-15

COMPREHENSIVE USER MANUAL

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1. INTRODUCTION & OVERVIEW

The Louis Martin F-Series represents a comprehensive range of professional passive loudspeaker systems engineered for demanding sound reinforcement environments. Spanning four models — F-8, F-10, F-12, and F-15 — the series delivers exceptional acoustic performance across a wide variety of application scenarios, from intimate lounge installations to high-energy live concert setups.

All F-Series cabinets share a common design philosophy centred on uncoloured, transparent sound reproduction. Each system employs a two-way passive crossover configuration, combining a high-power low-frequency driver with a precision compression HF driver housed in a 90°(H) × 50°(V) constant-directivity horn. Cabinets are constructed from high-density multi-layer plywood, ensuring both acoustic integrity and physical durability under continuous professional use.

Series Highlights

- Four models covering 8", 10", 12", and 15" low-frequency transducer sizes
- Two-way passive crossover design for plug-and-play simplicity
- Consistent 90°(H) × 50°(V) dispersion pattern across all models
- High-density multi-layer plywood cabinet construction
- Nominal 8 Ω impedance — compatible with standard professional amplifiers
- Flat frequency response for uncoloured, transparent audio reproduction
- Suitable for both permanent installation and portable touring applications

Intended Applications

The F-Series is engineered for a broad spectrum of professional audio environments including: lounges, bars and pubs, restaurants, multi-functional rooms, gymnasiums, classrooms, night clubs, live portable setups, and main front-of-house configurations.

■ All four models share identical connector types and mounting hardware philosophy, simplifying mixed deployments and system expansion.

2. SAFETY INSTRUCTIONS

■ READ ALL SAFETY INSTRUCTIONS CAREFULLY BEFORE OPERATING ANY F-SERIES LOUDSPEAKER SYSTEM. Failure to follow these instructions may result in personal injury, property damage, or voiding of the warranty.

Flying & Suspension

When flying speakers, appropriate rigging hardware and certified suspension techniques must be employed. All flying apparatus must conform to applicable local safety standards and load ratings. Never exceed the rated load capacity of any rigging component.

Regular Inspection

All speaker hardware, rigging points, brackets, and flying apparatus must be regularly inspected for any sign of visual or mechanical failure before each use. Immediately remove from service any component showing signs of wear, corrosion, or deformation.

Audience Distance

Maintain adequate distance between loudspeakers and the audience. Refer to your local Health and Safety authority for guidance on minimum separation distances for loudspeaker systems at various SPL levels.

Controller Programming

The Louis Martin digital controller must only be programmed by an authorised technician. Do not operate any F-Series loudspeaker without the correct controller and DSP preset installed. Operating the system without the correct preset may cause permanent driver damage and will void the warranty.

Environmental Protection

Protect all loudspeakers and associated electronics from exposure to humidity, condensation, water, rain, and extreme temperatures. These units are not rated for outdoor or weather-exposed use unless additional weatherproofing measures are in place.

Power Limits

Never exceed the rated power handling specifications for any F-Series model. Clipped or distorted amplifier output is the primary cause of driver failure and is not covered under warranty.

Ventilation

Ensure adequate airflow around amplifier equipment used with these systems. Do not obstruct ventilation openings on amplifiers or signal processing equipment.

Children & Unqualified Personnel

Keep children away from rigged loudspeaker systems. Installation and rigging should only be performed by qualified personnel with appropriate training and equipment.

3. F-8 LOUDSPEAKER SYSTEM

3.1 Product Overview

The F-8 is an 8-inch two-way passive loudspeaker system designed for compact installation and portable sound reinforcement. Its 8" (203 mm) low-frequency driver is paired with a 1.34" (34 mm) high-frequency compression driver, delivering full-range audio reproduction from a compact enclosure.

Featuring a passive crossover optimised for linearity, the F-8 provides flat frequency response from 70 Hz to 19 kHz, making it ideal for foreground music reproduction in restaurants, lounges, bars, classrooms, and gymnasiums. The compact cabinet dimensions allow multiple mounting configurations including wall brackets and speaker stands, while the 90°(H) × 50°(V) horn pattern ensures controlled, even coverage across the listening area.

Recommended Applications

- Lounge
- Pub / Bar
- Multi-functional room
- Gym / Fitness Centres
- Classroom

3.2 Technical Specifications

SPECIFICATION	VALUE
System Type	Two-way full-range passive loudspeaker
LF Driver	1 × 8" (203 mm) LF Driver
HF Driver	1 × 1.34" (34 mm) HF Driver
Frequency Response	70 Hz – 19 kHz (–10 dB)
RMS Power Handling	100 W
Program Power	200 W
Peak Power	400 W
Sensitivity	93 dB (1 W / 1 m)
Max SPL – Continuous	115 dB (1 m)
Max SPL – Peak	118 dB (1 m)
Nominal Impedance	8 Ω (typical)
Crossover	Passive, optimised for linearity
Dispersion Pattern	90°(H) × 50°(V)
Cabinet Material	High-density multi-layer plywood
Cabinet Dimensions	435 × 299 × 269 mm (H × W × D)

Weight

12 kg / 26.4 lbs

3.3 Product Views & Cabinet Dimensions

*Front View**Side View**Perspective View**Rear View*

Cabinet dimensions (H × W × D): **435 × 299 × 269 mm (H × W × D)** | Weight: **12 kg / 26.4 lbs**

■ All dimensions measured externally including grille. Mounting bracket and stand adaptor dimensions are not included in the figures above.

4. F-10 LOUDSPEAKER SYSTEM

4.1 Product Overview

The F-10 steps up to a 10" (254 mm) low-frequency driver fitted with a large 2.5" voice coil, providing significantly greater power handling (300 W RMS) and output capability compared to the F-8. This makes it suitable for medium-sized venues requiring higher SPL levels and extended low-frequency reproduction down to 60 Hz.

The F-10 retains the same passive crossover philosophy and 90°(H) × 50°(V) constant-directivity horn as the rest of the F-Series, ensuring consistent coverage patterns and easy system integration. At 96 dB sensitivity and a maximum continuous SPL of 123 dB, the F-10 delivers substantial headroom for demanding music playback environments including busy bars, multi-functional halls, and gymnasium sound systems.

Recommended Applications

- Lounge
- Pub / Bar
- Multi-functional room
- Gym / Fitness Centres
- Classroom

4.2 Technical Specifications

SPECIFICATION	VALUE
System Type	Two-way full-range passive loudspeaker
LF Driver	1 × 10" (254 mm) LF Driver with 2.5" Voice Coil
HF Driver	1 × 1.34" (34 mm) HF Driver
Frequency Response	60 Hz – 19 kHz (–10 dB)
RMS Power Handling	300 W
Program Power	600 W
Peak Power	1200 W
Sensitivity	96 dB (1 W / 1 m)
Max SPL – Continuous	123 dB (1 m)
Max SPL – Peak	126 dB (1 m)
Nominal Impedance	8 Ω (typical)
Crossover	Passive, optimised for linearity
Dispersion Pattern	90°(H) × 50°(V)
Cabinet Material	High-density multi-layer plywood
Cabinet Dimensions	500 × 351 × 317 mm (H × W × D)

Weight

17 kg / 37.4 lbs

4.3 Product Views & Cabinet Dimensions

*Front View**Side View**Perspective View**Rear View*

Cabinet dimensions (H × W × D): **500 × 351 × 317 mm (H × W × D)** | Weight: **17 kg / 37.4 lbs**

■ All dimensions measured externally including grille. Mounting bracket and stand adaptor dimensions are not included in the figures above.

5. F-12 LOUDSPEAKER SYSTEM

5.1 Product Overview

The F-12 brings 12" (304 mm) low-frequency transducer performance to the F-Series lineup. Equipped with a large-format 3" voice coil, the LF driver handles 400 W RMS with ease, while the upgraded 1.45" (37 mm) HF compression driver maintains pristine high-frequency reproduction up to 19 kHz.

Designed for Main Front-of-House applications, night clubs, live portable touring, and larger multipurpose halls, the F-12 delivers an impressive 126 dB continuous SPL and 129 dB peak SPL at one metre. The extended low-end response down to 55 Hz provides noticeable bass impact, particularly beneficial in music playback and live performance scenarios where sub-assisted configurations are not deployed.

Recommended Applications

- Lounge
- Pub / Bar
- Multi-functional room
- Live portable setup
- Night Clubs

5.2 Technical Specifications

SPECIFICATION	VALUE
System Type	Two-way full-range passive loudspeaker
LF Driver	1× 12" (304 mm) LF Driver with 3" Voice Coil
HF Driver	1× 1.45" (37 mm) HF Driver
Frequency Response	55 Hz – 19 kHz (–10 dB)
RMS Power Handling	400 W
Program Power	800 W
Peak Power	1600 W
Sensitivity	97 dB (1 W / 1 m)
Max SPL – Continuous	126 dB (1 m)
Max SPL – Peak	129 dB (1 m)
Nominal Impedance	8 Ω (typical)
Crossover	Passive, optimised for linearity
Dispersion Pattern	90°(H) × 50°(V)
Cabinet Material	High-density multi-layer plywood
Cabinet Dimensions	600 × 398 × 365 mm (H × W × D)

Weight

25 kg / 55.1 lbs

5.3 Product Views & Cabinet Dimensions

*Front View**Side View**Perspective View**Rear View*

Cabinet dimensions (H × W × D): **600 × 398 × 365 mm (H × W × D)** | Weight: **25 kg / 55.1 lbs**

■ All dimensions measured externally including grille. Mounting bracket and stand adaptor dimensions are not included in the figures above.

6. F-15 LOUDSPEAKER SYSTEM

6.1 Product Overview

The F-15 is the flagship model of the F-Series, engineered for the most demanding professional sound reinforcement applications. Its 15" (381 mm) low-frequency transducer with 3" voice coil delivers 500 W RMS power handling, while the large-format 1.75" (44 mm) HF compression driver ensures exceptional high-frequency detail and projection.

The F-15's passive crossover network is engineered to maintain phase coherence and linearity across the entire frequency range, minimising phase distortion at the crossover point. With a sensitivity of 99 dB and a peak SPL capability of 132 dB at one metre, the F-15 provides the power and headroom required for large-scale live events, night clubs, main FoH applications, and high-SPL permanent installations.

Recommended Applications

- Lounge
- Pub / Bar
- Multi-functional room
- Live portable setup
- Night Clubs

6.2 Technical Specifications

SPECIFICATION	VALUE
System Type	Two-way full-range passive loudspeaker
LF Driver	1× 15" (381 mm) LF Driver with 3" Voice Coil
HF Driver	1× 1.75" (44 mm) HF Driver
Frequency Response	50 Hz – 19 kHz (–10 dB)
RMS Power Handling	500 W
Program Power	1000 W
Peak Power	2000 W
Sensitivity	99 dB (1 W / 1 m)
Max SPL – Continuous	129 dB (1 m)
Max SPL – Peak	132 dB (1 m)
Nominal Impedance	8 Ω (typical)
Crossover	Passive, optimised for linearity
Dispersion Pattern	90°(H) × 50°(V)
Cabinet Material	High-density multi-layer plywood
Cabinet Dimensions	690 × 470 × 443 mm (H × W × D)

Weight	30 kg / 66.1 lbs
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6.3 Product Views & Cabinet Dimensions



Front View



Side View



Perspective View



Rear View

Cabinet dimensions (H × W × D): **690 × 470 × 443 mm (H × W × D)** | Weight: **30 kg / 66.1 lbs**

■ All dimensions measured externally including grille. Mounting bracket and stand adaptor dimensions are not included in the figures above.

7. F-SERIES COMPARISON CHART

The table below provides a side-by-side technical comparison of all four F-Series models to assist with system design and model selection.

SPECIFICATION	F-8	F-10	F-12	F-15
LF Driver	8" / 203 mm	10" / 254 mm	12" / 304 mm	15" / 381 mm
HF Driver	1.34" / 34 mm	1.34" / 34 mm	1.45" / 37 mm	1.75" / 44 mm
Voice Coil (LF)	—	2.5"	3"	3"
Frequency Response	70–19k Hz	60–19k Hz	55–19k Hz	50–19k Hz
RMS Power	100 W	300 W	400 W	500 W
Peak Power	400 W	1200 W	1600 W	2000 W
Sensitivity	93 dB	96 dB	97 dB	99 dB
Max SPL Continuous	115 dB	123 dB	126 dB	129 dB
Max SPL Peak	118 dB	126 dB	129 dB	132 dB
Impedance	8 Ω	8 Ω	8 Ω	8 Ω
Dispersion H×V	90°×50°	90°×50°	90°×50°	90°×50°
H × W × D (mm)	435×299×269	500×351×317	600×398×365	690×470×443
Weight	12 kg	17 kg	25 kg	30 kg

8. INSTALLATION GUIDELINES

Pre-Installation Planning

- Survey the venue acoustics prior to equipment placement. Identify reflective surfaces, parallel walls, and potential feedback paths.
- Calculate the required SPL at the furthest listening position using the speaker's sensitivity rating and the inverse-square law (SPL decreases by approximately 6 dB for each doubling of distance).
- Determine the optimal speaker placement height and angle for even audience coverage using the 90°(H) × 50°(V) dispersion pattern.
- Ensure all structural mounting points (walls, ceiling beams, rigging grids) are rated for the dynamic loads involved, including safety factors.

Mounting Configurations

- **STAND MOUNT:** All F-Series models feature a standard 35 mm pole socket on the bottom panel, allowing direct mounting on speaker stands and pole-mounted subwoofers.
- **WALL BRACKET:** Use appropriately rated wall-mount brackets (sold separately). Ensure all fixings penetrate into structural framing and are rated for the speaker's weight plus a minimum 4× safety factor.
- **FLOWN / RIGGED:** Use certified rigging hardware only. Attach to the factory-installed suspension points on the cabinet. All rigging must comply with EN 17206 or applicable local standards. A secondary safety wire must always be employed.
- **FLOOR MONITOR:** F-Series cabinets may be used as floor monitors when tilted to the appropriate angle using a dedicated wedge frame.

Cabling & Connectivity

- Use high-quality speaker cable with minimum cross-sectional area of 2.5 mm² for runs up to 10 metres, and 4 mm² for longer runs to minimise resistive losses.
- All F-Series models use NL4 / NL2 Neutrik Speakon connectors on the rear panel. Pin assignment: +1 = positive, -1 = negative.
- Daisy-chain connections (parallel wiring) are possible but will halve the impedance presented to the amplifier. Ensure the resulting combined impedance is within the amplifier's rated output range.
- Label all cables clearly at both ends and route them to avoid trip hazards, hot surfaces, and sources of electromagnetic interference (motors, fluorescent lights).

Aiming & Coverage

- Aim each cabinet to cover the intended listening zone. The -6 dB point of the 90°(H) dispersion pattern is at ±45° from the on-axis centre line.
- For wide-audience coverage, deploy multiple cabinets in a cluster or distributed arrangement. Use signal delay at distributed speakers to maintain speech intelligibility and a coherent stereo image.
- Avoid aiming speakers directly at microphones or reflective surfaces to minimise feedback and colouration.

9. AMPLIFIER PAIRING & SYSTEM SETUP

Selecting the correct amplifier is critical to achieving optimal performance and protecting your F-Series investment. The guidelines below will help you choose and configure the right amplification for each model.

Recommended Amplifier Power

As a general rule, amplifier RMS output power should be 1.5× to 2× the loudspeaker's RMS power rating when measured at 8 Ω. This headroom ensures the amplifier operates cleanly without clipping, which is the primary cause of HF driver failure.

MODEL	RMS RATING	RECOMMENDED AMP (8 Ω)	PEAK HANDLING
F-8	100 W	150 W – 200 W	400 W
F-10	300 W	450 W – 600 W	1200 W
F-12	400 W	600 W – 800 W	1600 W
F-15	500 W	750 W – 1000 W	2000 W

DSP Controller Setup

- Always use the Louis Martin digital controller with the correct F-Series preset loaded. Operating without the correct DSP preset removes crossover and driver-protection filters.
- Set high-pass and low-pass filter points as specified in the controller's preset. Do not manually override crossover frequencies unless directed by Louis Martin technical support.
- Apply appropriate gain structure from source to amplifier to maximise dynamic range and minimise the noise floor.
- Where available, enable amplifier clip limiters and thermal protection circuits.

■ **CLIPPING WARNING:** Clipped or distorted amplifier output generates high levels of odd-order harmonic energy in the HF band. This is the primary cause of compression driver failure and is NOT covered under warranty. Always ensure adequate amplifier headroom.

10. MAINTENANCE & CARE

Routine Inspection

- Inspect cabinet exterior for cracks, dents, or damage before each use.
- Check all rigging hardware, mounting brackets, and safety cables for signs of wear or deformation.
- Inspect Speakon connectors and cable connectors for bent pins, debris, or corrosion.
- Verify that grilles are firmly seated and show no significant damage that could affect driver protection.

Cleaning

- Clean cabinet surfaces with a slightly damp cloth. Avoid saturating the surface with liquid.
- Use a soft brush to remove dust accumulation from the grille cloth.
- Do not use solvents, abrasive cleaners, or chemical sprays on any surface of the cabinet.
- Keep HF horn mouths free from dust and debris — accumulated particulate can affect high-frequency dispersion.

Driver Care

- Avoid operating loudspeakers at excessive power levels for prolonged periods. Voice coil overheating is cumulative and causes irreversible damage.
- If distortion or rattling is detected during operation, immediately reduce level and investigate the source before continuing.
- Never probe or touch driver cones or diaphragms with fingers or tools.

Storage

- Store in a dry, temperature-stable environment when not in use.
- Use padded flight cases for transportation to protect against impact damage.
- If stacking cabinets, ensure weight distribution is even and cabinets are secured against movement.
- Allow cabinets that have been exposed to cold temperatures to acclimatise before applying power.

11. TROUBLESHOOTING

SYMPTOM	POSSIBLE CAUSE	RECOMMENDED ACTION
No sound output	No signal from amplifier; disconnected cable; failed amplifier channel	Check signal chain from source to amplifier. Test with known-good cable. Verify amplifier output.
Distorted sound	Amplifier clipping; incorrect DSP preset; damaged driver	Reduce gain. Verify DSP preset is loaded correctly. Inspect driver for physical damage.
Low-frequency dropout / thin sound	Single driver wired out of phase; missing LF driver connection	Check Speakon pin polarity. Verify both drivers are producing output individually.
Rattling / buzzing	Loose grille; foreign object in cabinet; damaged driver cone	Reseat or repair grille. Inspect interior of cabinet. Check driver cone for tears.
High-frequency loss	Blown HF compression driver; amplifier clipping; incorrect DSP	Test HF driver output in isolation. Verify amplifier is clean. Check DSP preset.
Hum or noise	Ground loop; poor cable shielding; nearby interference source	Use balanced XLR signal connections. Check cable routing away from power cables. Verify grounding.
Cabinet overheating	Excessive power for prolonged period; ambient temperature too high	Reduce drive level. Ensure adequate ventilation. Allow cooling period before resuming.
One cabinet significantly louder than another	Different gain levels; different cable lengths causing impedance variation	Match gain settings and cable lengths. Verify all connectors are fully seated.

■ If a fault cannot be resolved using the guidance above, contact your authorised Louis Martin distributor or service centre. Do not open cabinet enclosures or attempt driver replacement without proper training, as this may void the warranty.

12. WARRANTY INFORMATION

Warranty Coverage

Louis Martin loudspeakers and associated electronics are covered against defects in workmanship and materials for a period of two (2) years from the original date of purchase, subject to the conditions outlined below.

What Is Covered

- Manufacturing defects in materials or workmanship present at the time of purchase
- Electronic component failures under normal use conditions
- Cabinet structural failures resulting from manufacturing defects

What Is NOT Covered

■ The warranty does not cover damage resulting from user abuse, accidents, unauthorised modifications, incorrect controller configuration, operation outside rated specifications, physical impact damage, water ingress, or normal wear and tear.

- Driver damage caused by amplifier clipping or excessive power
- Physical damage from dropping, impact, or improper transport
- Water or moisture ingress damage
- Damage resulting from operation without the correct Louis Martin DSP controller preset
- Normal cosmetic wear on cabinet surfaces and grilles
- Items purchased from unauthorised resellers

Warranty Claim Procedure

- Contact your authorised Louis Martin distributor or service centre with proof of purchase and a description of the fault.
- Do not return goods without first obtaining a Return Material Authorisation (RMA) number.
- Pack the defective item securely in its original packaging (or equivalent protective packaging). Inadequately packed items that sustain transit damage may have the warranty claim refused.
- Ship the item pre-paid and insured to the authorised Louis Martin service centre. Return shipping of repaired / replaced items is covered by Louis Martin.
- Repaired or replaced items will be returned within a reasonable timeframe subject to parts availability.

Contact Information

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